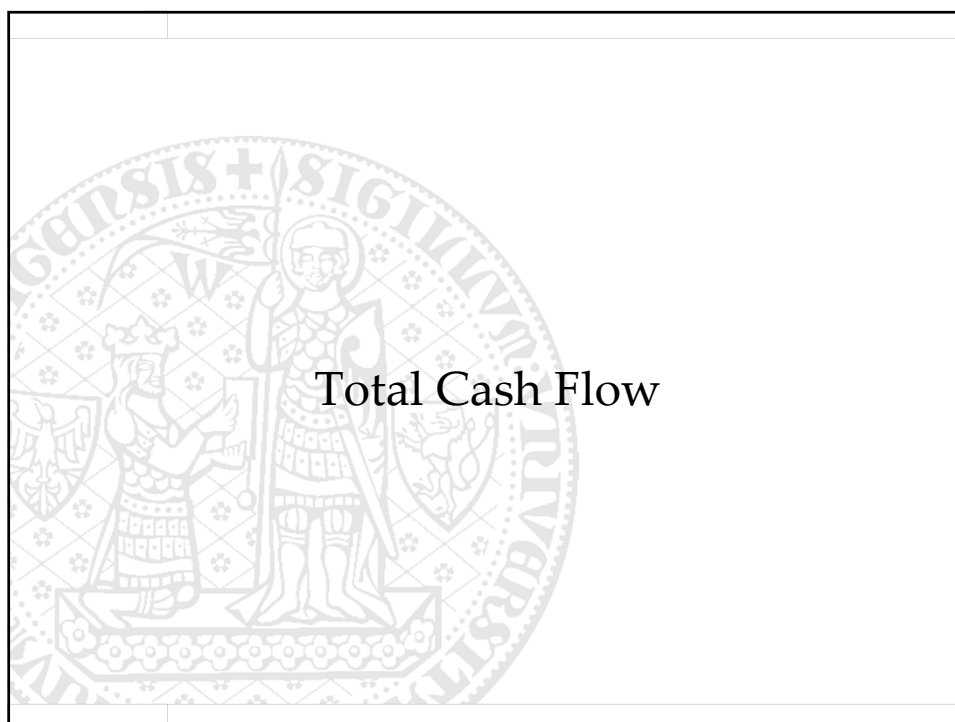
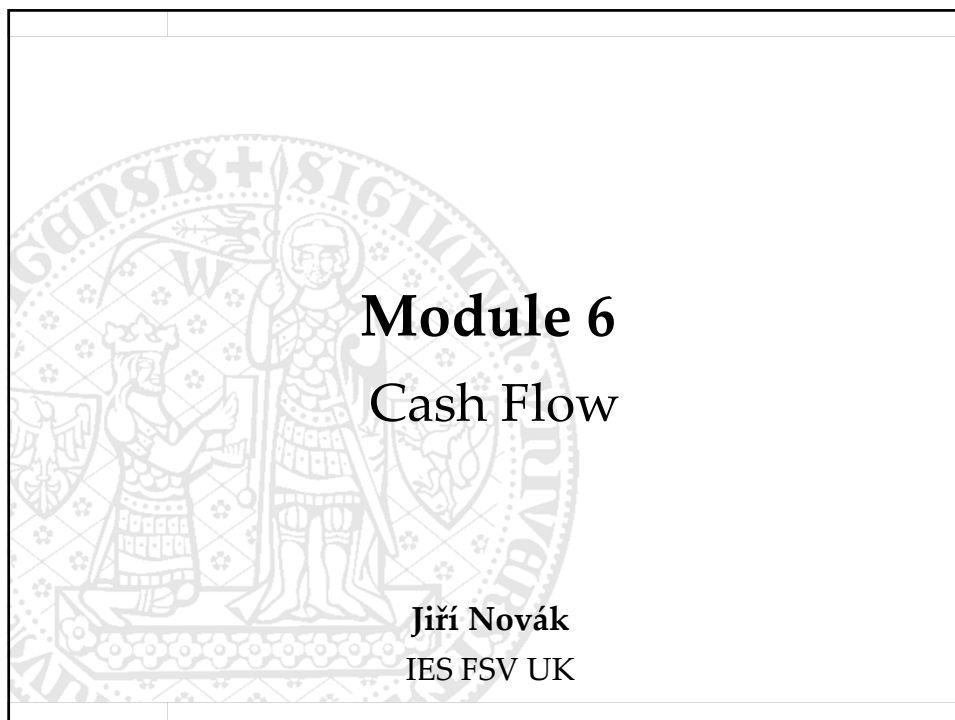






Statement of Cash Flow

- **Contents**
 - summarizes cash *receipts* and cash *outlays* that occur during fiscal year
- **Classification**
 - classified into *operating*, *investing*, and *financing*

Statement of Cash Flow
cash flow from operating activities
+/- cash flow from investing activities
+/- cash flow from financing activities
total cash flow

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Statement of Cash Flow

- **Questions**
 1. What is the simplest “shoe box” SCF?
 2. Is SCF more similar to IS or BS? Why?
 3. Is SCF related to the other statement? How?
 4. Where can we find total cash flow without using SCF?

- 4 -



Total Cash Flow



- **Change in Cash**

$$CF = \text{cash}_t - \text{cash}_{t-1} = \Delta \text{cash}$$
 - total cash flow is equal to change in cash “stock” on BS over fiscal year
- **Offsetting Changes**
 - to keep BS in balance cash level changes must be accompanied by changes in *As*, *Li*, or *Eq*
 - SCF explains BS changes in cash by changes in other accounts on BS

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Total Cash Flow

- **Liabilities**
 - increase in liabilities ~ cash inflow

Balance Sheet	
<i>cash</i>	<i>liabilities</i>
<i>non-cash assets</i>	<i>equity</i>

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Total Cash Flow

- **Equity**
 - increase in equity ~ cash inflow

Balance Sheet	
<i>cash</i>	<i>liabilities</i>
<i>non-cash assets</i>	<i>equity</i>

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Total Cash Flow

- **Assets**
 - increase in non-cash assets ~ cash outflow

Balance Sheet	
<i>cash</i>	<i>liabilities</i>
<i>cash outflow</i>	
<i>non-cash assets</i>	<i>equity</i>

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Total Cash Flow

- **Balance Sheet Equation**

$$As_t = Li_t + Eq_t$$

$$As_{t-1} = Li_{t-1} + Eq_{t-1}$$

$$\Delta As = \Delta Li + \Delta Eq$$

- **Change in Cash**

$$\begin{aligned} \overbrace{\Delta \text{cash} + \Delta \text{other } As}^{\Delta \text{cash}} &= \Delta Eq + \Delta Li \\ &= \Delta Eq + \Delta Li - \Delta \text{other } As \end{aligned}$$

- **Offsetting Changes**

- when reconciling cash flow and income:

$$\uparrow \text{cash} \quad \sim \uparrow Eq / \uparrow Li / \downarrow \text{other } As$$

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Cash Flow Components



Classification



- **Sections**
 - SCF has 3 sections: *operating*, *investing*, *financing*
- **Business Activities**
 - defined based on their purpose:
 - *financing* – acquisition of *capital* (equity, debt), depositing excess liquidity
 - *investing* – acquisition of assets of *long-term* benefit (fixed assets)
 - *operating* – management of ordinary operations (all other)

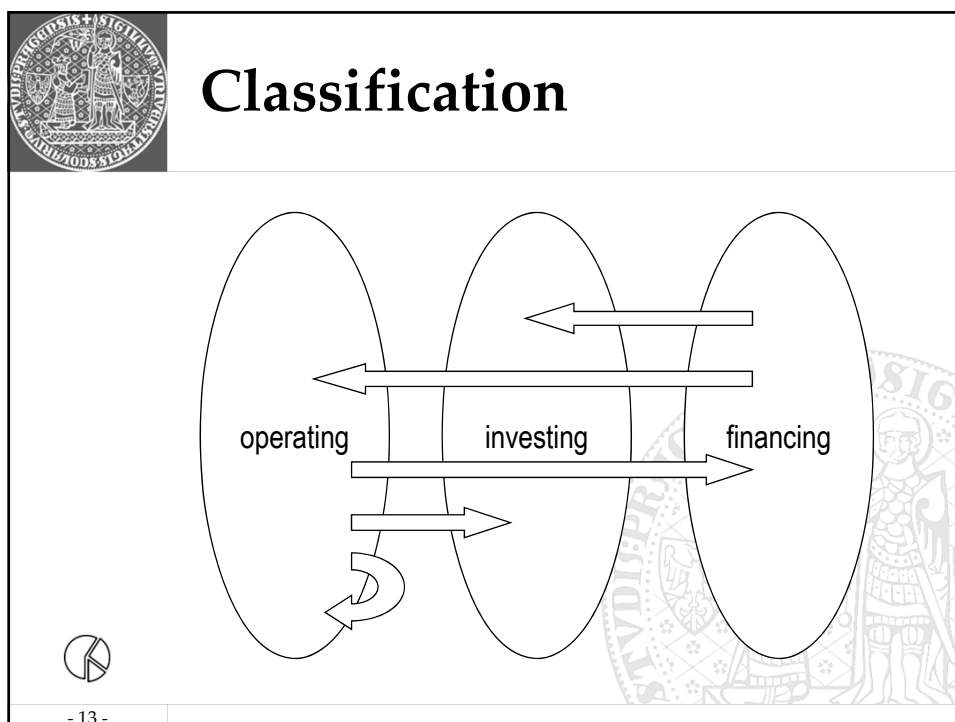
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Classification

- **Financing**
 - issuance & retirement of debt
 - issuance & retirement of equity
 - payment of dividends, repurchase of stocks
- **Investing**
 - acquisition/sale of tangible fixed assets
 - acquisition/sale of intangible assets
 - acquisition/sale of subsidiary/share in other firm
- **Operating**
 - all remaining activities ~ *Rv* & *Ex* in IS

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Classification

- **Criteria**
 - to attribute changes in *As* and *Li* to three SCF sections, we need to distinguish between:
 - *operating* – *financing* – distinguish *CFO* & *CFF*
 - *short-term* – *long-term* – distinguish *CFO* & *CFI*

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Classification

Balance Sheet	
<i>cash</i>	<i>COL</i>
<i>COA</i>	<i>CFL</i>
<i>CFA</i>	<i>LTFL</i>
<i>XA</i>	<i>CC</i>
	<i>ARE</i>

Income Statement	
<i>operating expenses</i>	<i>revenues</i>
<i>depreciation</i>	
<i>financial exp.</i>	
<i>earnings</i>	

- financing
- investing
- operating

}
merged in IS

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Formula

- **Clean Surplus Relationship**
 - accumulated retained earnings (*ARE*):
 - increase by earnings (*NI*),
 - decrease by dividends (*Dv*)
 - $ARE_t = ARE_{t-1} + NI_t - Dv_t$
 - $\Delta ARE_t = NI_t - Dv_t$

NI (income) →

beginning balance

ARE (Eq)

→ *Dv (cash)*

ending balance

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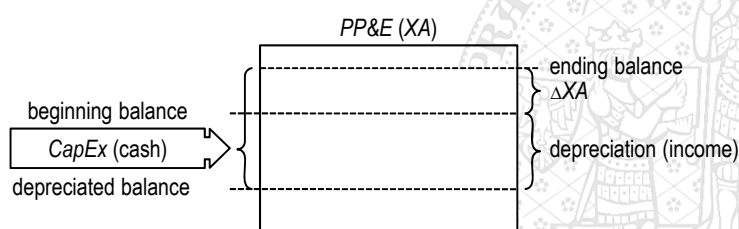


Formula

Fixed Assets

- capital expenditures (*CapEx*) cash paid for XA:
 - replacements of depreciated XA (*Dp*)
 - extensions – change in net fixed assets (ΔXA)

$$\begin{aligned} CapEx &= Dp + \Delta XA \\ \Delta XA &= CapEx - Dp \end{aligned}$$



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Formula

Other Assets

$$\Delta \text{other As} = \Delta \text{COA} + \Delta \text{CFA} + \Delta XA$$

$$\Delta \text{other As} = \Delta \text{COA} + \Delta \text{CFA} + \overbrace{CapEx - Dp}$$

Liabilities

$$\Delta Li = \Delta COL + \Delta CFL + \Delta LTFL$$

Equity

$$\Delta Eq = \Delta CC + \Delta ARE$$

$$\Delta Eq = \Delta CC + \overbrace{NI - Dv}$$

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Formula

▪ **Cash Flow**

$$\Delta \text{cash} = \Delta E_q + \Delta Li - \Delta \text{other As}$$

$$\Delta \text{cash} = \underbrace{\Delta CC + NI - Dv}_{\text{... CFO}} + \underbrace{\Delta COL + \Delta CFL + \Delta LTFL}_{\text{... CFI}} - \underbrace{\Delta COA - \Delta CFA - CapEx + Dp}_{\text{... CFF}}$$

$$\Delta \text{cash} = NI + Dp - \Delta COA + \Delta COL - \dots \text{CFO}$$

$$- CapEx + \dots \text{CFI}$$

$$+ \Delta CC - Dv + \Delta LTFL + \Delta CFL - \Delta CFA \dots \text{CFF}$$

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Indirect Method



Indirect Statement of Cash Flow	
	net income
+	depreciation
-	increase in current assets
+	increase in current operating liabilities
	cash flow from operations
-	capital expenditures
-	financial investments
	cash flow from investing
+(-)	issuance (retirement) of debt
+(-)	issuance (repurchase) of common stock
-	dividends paid out
	cash flow from financing



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Cash Flow

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Extra – Classification



Classification

- **Cost Benefit Mismatch**
 - (intentional) *mismatch* in classification between cash generated and sources invested
 - acquisition of fixed assets is *CFI*, when assets generate revenues and cash inflows it is *CFO*
 - investment into securities is *CFI*, cash received from interest and dividend revenues is *CFO*
 - note that maintaining operations requires cash outflows in *CFO*, but also investments in *CFI*

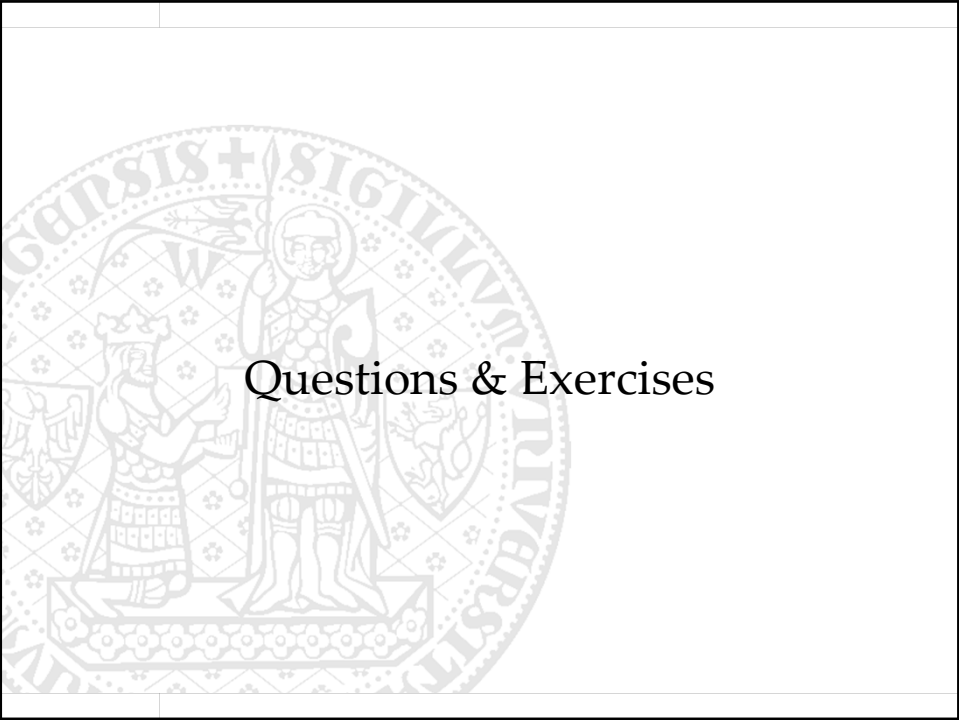
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Classification

- **US GAAP vs. IFRS**
 - historically classification of interest expense differed between IFRS and US GAAP
 - US GAAP – issue/redemption of debt and equity is *CFF*, interest payment is *CFO*, but dividends paid out to shareholders is *CFF*
 - IFRS past – issue/redemption of debt & equity, interest & dividends payments are *CFF*
 - IFRS now – allows classification of both interest & dividend paid as either *CFO* or *CFF*

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Questions & Exercises



Questions



▪ Timing Sequence

- What is *most common* timing sequence of positive and negative income and cash flows related to *operating* activities?

	negative	positive
cash	outflow	inflow
income	expense	revenue

- Different timing of cash outflows and inflows implies that firms need ... ?



Questions

- **Cash Paid to Suppliers**

- One of the examples used in the lecture shows how to find the amount of cash paid to suppliers of raw materials in case the firm pays for inventories immediately in cash.
- Relaxing the assumption of all inventories are purchased for cash, how can we determine the cash paid for inventory if we only have the Income Statement and the Balance Sheet?

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Questions



- **Sections of IS and SCF**

- Why does the Statement of Cash Flow have three sections while the Income Statement just two?

- **Relevance vs. Reliability**

- Assess relative relevance and reliability of earnings and cash flows?

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Exercise

- **Required**

- using the end-of-the-year BS and IS (provided) prepare the indirect statement of cash flow for the company making transactions from the Accounting Cycle Exercise (Lecture 2); explain briefly the individual steps you make
- optionally, you may also prepare *CFO* using the direct method

- **Understanding**

- go through the individual transactions and identify their position on SCF

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Exercise

- **Preparation**

- calculate the changes of all the balance sheet accounts (but for cash)
- list the changes as inflows or outflows
- categorize the flows by operating financing and investing activities

- **Verification**

- make sure that the total cash flow is equal to the change in cash on the balance sheet

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Exercise

Balance Sheet

	2001	2000		2001	2000
cash	710	0	accounts payable	200	0
raw materials	100	0	salaries payable	110	0
finished goods	0	0	taxes payable	80	0
accounts receivable	150	0	current liabilities	390	0
current assets	960	0	bank loan	400	0
machinery	500	0	long-term liabilities	400	0
accum. depreciation	-50	0	contributed capital	600	0
fixed assets	450	0	retained earnings	20	0
			equity	620	0
total assets	1410	0	total equity & liabilities	1410	0

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Exercise

Income Statement

	2001
net sales	500
cost of goods sold	-100
gross profit	400
salaries	-110
depreciation	-50
earnings before interest and taxes	240
interest expense	-40
earnings before income taxes	200
income taxes	-80
net income	120

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Exercise

▪ **Prepare indirect SCF for the company making transactions from Exercise 2.**

- 1) owners start up the company with €600 in cash
- 2) it obtains a bank loan of €400
- 3) it purchases for cash a machine for €500 that is going to be depreciated over 10 years
- 4) it purchases raw material for €200 on account (i.e. on credit)
- 5) half of the raw material is used up to make finished goods
- 6) salaries to central management of €110 accrue (to be paid out next period)
- 7) first year depreciation for the machine is recognized
- 8) interest on loan paid by cash; interest rate is 10%
- 9) all finished goods are sold for €500 on credit (! !)
- 10) receivables worth €350 are collected
- 11) profit is taxed at 40%
- 12) dividends of €100 are paid out from after tax profit

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